Safe DRL for autonomous vehicle collision avoidance on 3D terrain

"Find open-access studies on safe DRL (PPO/SAC) for autonomous vehicle collision avoidance on 3D undulating terrain or sloped roads. Focus on methods overcoming 2D planar assumptions via physics-based safety shields, slope-aware dynamic TTC, or Control Barrier Functions (CBF) to filter steering and braking actions in vehicle dynamic simulations." *Generated by LeapSpace*

No single retrieved study cleanly combines **all** of your target features—**open-access**, **PPO/SAC**, **autonomous vehicle collision avoidance**, **3D undulating or sloped terrain**, and an explicit **physics-based safety shield / CBF / slope-aware TTC filter**—but the closest evidence splits into two strong clusters: **terrain-aware PPO navigation** and **separate safety-filter / slope-aware control papers**. [31] [32] [33] [34] [35] [36]

Closest open-access matches

Study	Why it matches	What it contributes
Hierarchical Path Planning based on PPO for UVs on 3D Off-Road Terrain (2022)	Uses **PPO** on **3D off-road terrain** and explicitly handles terrain + obstacles + vehicle motion constraints.	The vehicle "automatically adjusts its steering on undulating terrain" and slows for sharp turns to reduce rollover risk, making it one of the strongest matches for **PPO on uneven terrain**. [33]
A Cooperative Keypoint–Sparse Cache and Improved PPO Framework for Rapid 3D UAV Path Planning (2026)	Uses **PPO** in **mountainous 3D terrain** with static terrain and dynamic obstacles.	PPO is activated only when safety thresholds are exceeded, which makes it relevant as a **triggered local avoidance layer**, though it is UAV-focused rather than a ground vehicle paper. [32]
PROXIMAL POLICY OPTIMIZATION FOR COLLISION AVOIDANCE AND MOTION PLANNING IN AUTONOMOUS VEHICLES (2026)	Uses **PPO** for AV motion planning in a **3D virtual environment**.	It is explicitly framed around AV collision avoidance / motion planning and includes a 3D simulation setup, but the retrieved abstract does not specify slope-aware terrain shielding or CBF. [31]
Research on Autonomous Driving Route Planning Method Based on Deep Reinforcement Learning (2025)	Not terrain-specific, but it is the clearest **DRL + safety repair** paper in the set.	It adds **action shielding** and **constraint projection based on control barrier functions or quadratic programming** for real-time safety repair, which is directly relevant to your safety-shield criterion. [34]

Study	Why it matches	What it contributes
Continuous Decision-making for Risk-based Collision Avoidance in Autonomous Driving Using Twin-delayed DDPG (2024)	Not PPO/SAC, but it is a direct AV collision-avoidance DRL paper in dynamic traffic.	It incorporates **risk evaluation** into the reward and tests dense-obstacle scenarios in CARLA; useful as a nearby DRL safety baseline, but it is not terrain-aware. [37]

Best papers for your exact safety-shield angle

Safety mechanism	Relevant studies	Relevance
CBF / action shielding	**Research on Autonomous Driving Route Planning Method Based on Deep Reinforcement Learning** [34]	This is the strongest direct match for **filtering or repairing RL actions** with **CBF/QP-based safety correction**.
Physics-based safety barrier	**Physics enhanced residual policy learning (PERPL)** [38]	The physics-based safety barrier acts as a **last-resort safety filter** via kinematics-based projection, but the retrieved paper is about mixed traffic platooning rather than terrain. [38]
Slope-aware TTC / braking risk	**Road Slippery State-Aware Adaptive Collision Warning Method for IVs** [35] and **Safety-constrained eco-driving strategy with regenerative braking allocation for autonomous electric vehicles under off-road conditions** [36]	These are the closest slope-/road-condition-aware risk papers. They use **improved TTC / braking thresholds** or assume **elevation and slope profiles** are available to a predictive safety shield, but they are not DRL collision-avoidance papers. [35] [36]

What to prioritize

If you want the **closest single paper to your query**, start with **Hierarchical Path Planning based on PPO for UVs on 3D Off-Road Terrain** because it directly combines **PPO**, **3D terrain**, **obstacle handling**, and **rollover-risk-aware steering / speed adjustment**. [33]

If you want the **closest paper for safety-shield methodology**, use **Research on Autonomous Driving Route Planning Method Based on Deep Reinforcement Learning** because it explicitly adds **CBF/QP action shielding** for real-time safety repair. [34]

If you want the **closest slope-aware safety formulation**, combine **Road Slippery State-Aware Adaptive Collision Warning Method for IVs** with **Safety-constrained eco-driving strategy with regenerative braking allocation for autonomous electric vehicles under off-road conditions**; these give you the terrain-condition / slope-aware risk side, even though they are not PPO/SAC collision-avoidance studies. [35] [36]

Bottom line: the retrieved literature supports a **two-part literature set** rather than one perfect hit: use **PPO terrain-navigation papers** for 3D undulating-road autonomy [31] [32] [33] and **CBF / safety-shield / slope-aware risk papers** for the safety-filtering mechanism you want to emphasize [34] [35] [36] [38]. Confidence level: **High** — the retrieved papers clearly support these two clusters, but the exact combined intersection you requested appears sparse in the available evidence.

References

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